

```

SELECT
    e.Name as employee_Name,
    p.PName as project_Name
FROM employee e
join works_on w
    on e.SSN = w.ESSN
JOIN project p
    on w.PNumber = p.PNumber
ORDER by p.PName;

```

line 163, column 16, location 3771

No limit



employee_Name	project_Name
Ahmed Ali	AL Rabwah
Mona Hassan	AL Rabwah
Kamel Mohamed	AL Rabwah
Omar Ahmed	Alpha Project
Sara Mohamed	Beta Project
Ali Kamel	Sales System

orks_on

```
337  
338  
339  
340  
341 CREATE INDEX idx_employee_name  
342 ON Employee(Name);  
343  
344
```

 line 342, column 19, location 6557

Query 1 OK

```
SELECT
    D.DName AS Department_Name,
    P.PName AS Project_Name
FROM Department D
JOIN Project P
    ON D.DNumber = P.DNumber;
```

line 164, column 30, location 3811

No limit



Beautify



Run Current



Department_Name	Project_Name
-----------------	--------------

Research	AL Rabwah
----------	-----------

Research	Alpha Project
----------	---------------

Administration	Beta Project
----------------	--------------

Sales	Sales System
-------	--------------

```
269 -- _____transaction
270
271 START TRANSACTION;
272
273 INSERT INTO Employee
274 (SSN, Name, Address, Gender, Salary, BD, DNO, Superssn)
275 VALUES
276 (102006, 'Mostafa Ahmed', 'Cairo', 'M', 2200, '1994-05-15', 30, 102001);
277
278 COMMIT;
```



line 271, column 1, location 5387

No limit



Beautify



Run Current



Query 1

Query 2

Query 3

Query 3 OK

```
156  
157 INSERT INTO Employee  
158 (SSN, Name, Address, Gender, BD, DNO)  
159 VALUES  
160 (102660, 'Friend Name', 'Friend Address',  
161 'M', '2000-01-01', 30);|
```

 line 161, column 25, location 3809

No limit



Beautify



Run Current



Query 1 OK: 1 rows affected

```
346 -- CREATE INDEX idx_employee_department_salary
347 -- ON Employee(DNO, Salary);
348
349
350 SHOW INDEX FROM Employee;
```

line 350, column 26, location 6678

No limit Beautify Run Current

	Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	E
1	employee	0	PRIMARY	1	SSN	A	10	NULL	NULL	EMPTY	BTREE	EMPTY	EMPTY	YES	NU
2	employee	1	Superssn	1	Superssn	A	3	NULL	NULL	YES	BTREE	EMPTY	EMPTY	YES	NU
3	employee	1	idx_employee_name	1	Name	A	10	NULL	NULL	EMPTY	BTREE	EMPTY	EMPTY	YES	NU
4	employee	1	idx_employee_department_salary	1	DNO	A	3	NULL	NULL	YES	BTREE	EMPTY	EMPTY	YES	NU
5	employee	1	idx_employee_department_salary	2	Salary	A	9	NULL	NULL	YES	BTREE	EMPTY	EMPTY	YES	NU

```

SELECT
    Dep.*,
    E.Name AS Employee_Name
FROM Dependences Dep
JOIN Employee E
    ON Dep.ESSN = E.SSN;

```

line 164, column 25, location 3789

No limit



Beautify

ESSN	ID	Name	BD	Gender	Employee_Name
102001	1	Sara Ahmed	2018-01-20	F	Ahmed Ali
102004	1	Adam Sara	2015-08-10	M	Sara Mohamed
112233	1	Mariam Kamel	2010-02-10	F	Kamel Mohamed
112233	2	Omar Kamel	2012-04-15	M	Kamel Mohamed

```
SELECT
    PNumber,
    PName,
    Location
FROM Project
WHERE Location IN ('Cairo', 'Alex');
```

line 164, column 37, location 3775

No limit



Beautify



Run Current

PNumber	PName	Location
100	AL Rabwah	Cairo
101	Alpha Project	Cairo
102	Beta Project	Alex

```
292  
293  
294 START TRANSACTION;  
295  
296 UPDATE Employee  
297 SET Salary = Salary * 1.05;  
298  
299 ROLLBACK;
```



line 294, column 1, location 5824

No limit



Beautify



Run Current

Query 1

Query 2

Query 3

Query 3 OK

```
344  
345  
346 CREATE INDEX idx_employee_department_salary  
347 ON Employee(DNO, Salary);  
348
```



line 347, column 26, location 6641

No limit



Beautify



Query 1 OK

```
157
158
159 SELECT
160     P.PNumber AS Project_Number,
161     D.DName AS Department_Name,
162     SUBSTRING_INDEX(M.Name, ' ', -1) AS Manager_Last_Name,
163     M.Address AS Manager_Address,
164     M.BD AS Manager_Birthdate
165 FROM Project P
166 JOIN Department D
167     ON P.DNumber = D.DNumber
168 JOIN Employee M
169     ON D.ManagerSSN = M.SSN
170 WHERE P.Location = 'Cairo';
```

 line 166, column 18, location 3911

No limit



Beautify



Run Current



Project_Number	Department_Name	Manager_Last_Name	Manager_Address	Manager_Birthdate
----------------	-----------------	-------------------	-----------------	-------------------

```
-- DELIMITER //
```

```
-- CREATE PROCEDURE GetEmployeesByDepartment(IN DeptNo INT)
```

```
-- BEGIN
```

```
--     SELECT *
```

```
--     FROM Employee
```

```
--     WHERE DNO = DeptNo;
```

```
-- END //
```

```
-- DELIMITER ;
```

```
CALL GetEmployeesByDepartment(30);
```

line 182, column 35, location 4149

No limit



Beautify



Run Current



SSN	Name	Address	Gender	Salary	BD	DNO	Superssn
102001	Ahmed Ali	Cairo	M	1500.00	1990-03-15	30 →	112233 →
102002	Mona Hassan	Alex	F	1800.00	1992-07-20	30 →	112233 →
102005	Ali Kamel	Cairo	M	1200.00	1993-09-18	30 →	102001 →
102660	Friend Name	Friend Address	M	NULL	2000-01-01	30 →	NULL →
102672	Your Name	Your Address	M	1500.00	2000-01-01	30 →	112233 →

```

SELECT
  E.*,
  Dep.ID AS Dependent_ID,
  Dep.Name AS Dependent_Name,
  Dep.BD AS Dependent_BD,
  Dep.Gender AS Dependent_Gender
FROM Employee E
LEFT JOIN Dependences Dep
  ON E.SSN = Dep.FSSN

```

line 166, column 25, location 3888

No limit



Beautify



SSN	Name	Address	Gender	Salary	BD	DNO	Superssn	Dependent_ID	Dependent_Name	Dependent_BD	Dependent_Gender
102001	Ahmed Ali	Cairo	M	1500.00	1990-03-15	30 →	112233 →	1	Sara Ahmed	2018-01-20	F
102002	Mona Hassan	Alex	F	1800.00	1992-07-20	30 →	112233 →	NULL	NULL	NULL	NULL
102003	Omar Ahmed	Cairo	M	3000.00	1988-11-12	10 →	112233 →	NULL	NULL	NULL	NULL
102004	Sara Mohamed	Alex	F	2500.00	1995-01-25	20 →	112233 →	1	Adam Sara	2015-08-10	M
102005	Ali Kamel	Cairo	M	1200.00	1993-09-18	30 →	102001 →	NULL	NULL	NULL	NULL
112233	Kamel Mohamed	Cairo	M	5000.00	1980-05-10	10 →	NULL →	1	Mariam Kamel	2010-02-10	F
112233	Kamel Mohamed	Cairo	M	5000.00	1980-05-10	10 →	NULL →	2	Omar Kamel	2012-04-15	M

```
154
155 SELECT
156     P.PName AS Project_Name,
157     COALESCE(SUM(W.Hours), 0) AS Total_Hours
158 FROM Project P
159 LEFT JOIN Works_On W
160     ON P.PNumber = W.PNumber
161 GROUP BY P.PNumber, P.PName;
```



line 161, column 29, location 3849

No limit



Beautify



Run Current



	Project_Name	Total_Hours
1	AL Rabwah	40.00
2	Alpha Project	15.00
3	Beta Project	10.00
4	Sales System	25.00

```
3  
9 SELECT *  
9 FROM Project  
1 WHERE PName LIKE 'a%';  
2
```

line 161, column 23, location 3723

No limit



Beautify



PNumber	PName	Location	DNumber
100	AL Rabwah	Cairo	10
101	Alpha Project	Cairo	10

```
57  
58 SELECT DISTINCT  
59     SUBSTRING_INDEX(E.Name, ' ', -1) AS Manager_Last_Name  
60 FROM Employee E  
61 JOIN Department D  
62     ON E.SSN = D.ManagerSSN  
63 LEFT JOIN Dependences Dep  
64     ON E.SSN = Dep.ESSN  
65 WHERE Dep.ESSN IS NULL;
```

 line 165, column 24, location 3891

No limit



Beautify



Run Current



Manager_Last_Name

```

182 -- CALL GetEmployeeByDepartment(50);
183
184
185
186 -- DELIMITER //
187
188 -- CREATE PROCEDURE GetEmployeeBySSN(IN EmployeeSSN INT)
189 -- BEGIN
190
191 --     SELECT *
192 --     FROM Employee
193 --     WHERE SSN = EmployeeSSN;
194
195 -- END //
196
197 -- DELIMITER ;
198
199 CALL GetEmployeeBySSN(102003);

```



line 199, column 1, location 4354

No limit



Beautify



Run Current



	SSN	Name	Address	Gender	Salary	BD	DNO	Superssn
1	102003	Omar Ahmed	Cairo	M	3000.00	1988-11-12	10 →	112233 →

```
220
221 DELIMITER //
222
223 CREATE PROCEDURE CountEmployeesInDepartment(IN DeptNo INT)
224 BEGIN
225
226     SELECT COUNT(*) AS TotalEmployees
227     FROM Employee
228     WHERE DNO = DeptNo;
229
230 END //
231
232 DELIMITER ;
233
234 CALL CountEmployeesInDepartment(30);
235
```



line 221, column 1, location 4654

No limit



Beautify



Query 1

Query 2

TotalEmployees

1

5

```

SELECT
    D.DNumber AS Department_ID,
    D.DName AS Department_Name,
    E.SSN AS Manager_ID,
    E.Name AS Manager_Name
FROM Department D
JOIN Employee E
    ON D.ManagerSSN = E.SSN;

```

line 166, column 29, location 3869

No limit



Beautify



Run Current



Department_ID	Department_Name	Manager_ID	Manager_Name
30	Sales	102001	Ahmed Ali

```
153 -- (102004, 1, 'Adam Sara', '2015-08-10', 'M');
154
155
156 INSERT INTO Department
157 (DNumber, DName, ManagerSSN, ManagerStartDate)
158 VALUES
159 (100, 'IT', 112233, '2006-11-01');
```

 line 159, column 32, location 3782

No limit ▼

Beautify ▼

Run Current ▼

Query 1 OK: 1 rows affected

```
285  
286 INSERT INTO Employee  
287 (SSN, Name, Address, Gender, Salary, BD, DNO, Superssn)  
288 VALUES  
289 (102007, 'Ahmed Samir', 'Alex', 'M', 2500, '1991-08-20', 20, 112233);  
290  
291 ROLLBACK;
```



line 284, column 1, location 5609

No limit



Beautify



Run Current



Query 1

Query 2

Query 3

Query 3 OK

```
SELECT
    E.Name
FROM Employee as E
JOIN Employee as S
    ON E.superssn = S.SSN
WHERE S.Name = 'Kamel Mohamed';
```

line 162, column 12, location 3727

No limit



Beautify



Run Current



Name

Ahmed Ali

Mona Hassan

Omar Ahmed

Sara Mohamed

```
155
156 SELECT
157     Dep.Name AS Dependent_Name,
158     Dep.Gender
159 FROM Dependences Dep
160 JOIN Employee E
161     ON Dep.ESSN = E.SSN
162 WHERE Dep.Gender = 'F'
163 AND E.Gender = 'F'
164
165 UNION
166
167 SELECT
168     Dep.Name AS Dependent_Name,
169     Dep.Gender
170 FROM Dependences Dep
171 JOIN Employee E
172     ON Dep.ESSN = E.SSN
173 WHERE Dep.Gender = 'M'
174 AND E.Gender = 'M';|
```



line 174, column 20, location 4011

No limit



Beautify



Run Current



	Dependent_Name	Gender
--	----------------	--------

1	Omar Kamel	M
---	------------	---

```
SELECT E.Name
FROM Employee E
JOIN Works_On W
    ON E.SSN = W.ESSN
JOIN Project P
    ON W.PNumber = P.PNumber
WHERE E.DNO = 10
AND W.Hours >= 10
AND P.PName = 'AL Rabwah';
```

line 167, column 27, location 3858

Name

Kamel Mohamed

```
SELECT
  D.DName AS Department_Name,
  MAX(E.Salary) AS Maximum_Salary,
  MIN(E.Salary) AS Minimum_Salary,
  AVG(E.Salary) AS Average_Salary
FROM Department D
LEFT JOIN Employee E
  ON D.DNumber = E.DNO
GROUP BY D.DNumber, D.DName;
```

line 166, column 29, location 3924

No limit



Beautify



Run Current



Department_Name	Maximum_Salary	Minimum_Salary	Average_Salary
Research	5000.00	3000.00	4000.000000
Administration	2500.00	2500.00	2500.000000
Sales	1800.00	1200.00	1500.000000
IT	NULL	NULL	NULL

Indexes improve query performance by making searches faster, especially with WHERE, JOIN, and ORDER BY.

```
SELECT *  
FROM Employee  
WHERE DNO = 30  
AND Salary BETWEEN 1000 AND 2000;
```

line 162, column 34, location 3751

No limit



Beautify



Run

SSN	Name	Address	Gender	Salary	BD	DNO	Superssn
102001	Ahmed Ali	Cairo	M	1500.00	1990-03-15	30	112233
102002	Mona Hassan	Alex	F	1800.00	1992-07-20	30	112233
102005	Ali Kamel	Cairo	M	1200.00	1993-09-18	30	102001

```
154
155 SELECT D.*
156 FROM Department D
157 JOIN Employee E
158     ON D.DNumber = E.DNO
159 WHERE E.SSN = (
160     SELECT MIN(SSN)
161     FROM Employee
162 );
```



line 162, column 3, location 3802

No limit



Beautify



Run

	DNumber	DName	ManagerSSN	ManagerStartDate
--	---------	-------	------------	------------------

1	30	Sales	102001 →	2015-06-01
---	----	-------	----------	------------

57

58

59 SELECT DISTINCT E.*

60 FROM Employee E

61 JOIN Department D

62 ON E.SSN = D.ManagerSSN;



line 162, column 29, location 3762

No limit



Beautify



Run Current



	SSN	Name	Address	Gender	Salary	BD	DNO	Superssn
1	102001	Ahmed Ali	Cairo	M	1500.00	1990-03-15	30 →	112233 →

```
202
203
204
205 -- DELIMITER //
206
207 -- CREATE PROCEDURE IncreaseDepartmentSalary(IN DeptNo INT)
208 -- BEGIN
209
210 --     UPDATE Employee
211 --     SET Salary = Salary * 1.10
212 --     WHERE DNO = DeptNo;
213
214 -- END //
215
216 -- DELIMITER ;
217
218 -- CALL IncreaseDepartmentSalary(30);
219
220
```



line 220, column 1, location 4652

No limit



Beautify



Run Current

Query 1 OK: 4 rows affected

```
351  
352 DROP INDEX idx_employee_name  
353 ON Employee;
```



line 353, column 13, location 6727

No limit



Beautify

Query 1 OK

```

238
239 -- DELIMITER //
240
241 -- CREATE PROCEDURE GetHighestPaidEmployeePerDepartment()
242 -- BEGIN
243
244 --     SELECT E.*
245 --     FROM Employee E
246 --     JOIN
247 --     (
248 --         SELECT DNO, MAX(Salary) AS MaxSalary
249 --         FROM Employee
250 --         GROUP BY DNO
251 --     ) M
252 --     ON E.DNO = M.DNO
253 --     AND E.Salary = M.MaxSalary;
254
255 -- END //
256
257 -- DELIMITER ;
258
259
260 CALL GetHighestPaidEmployeePerDepartment();

```



line 260, column 44, location 5336

No limit



Beautify



Run Current



	SSN	Name	Address	Gender	Salary	BD	DNO	Superssn
1	102002	Mona Hassan	Alex	F	2178.00	1992-07-20	30 →	112233 →
2	102004	Sara Mohamed	Alex	F	2500.00	1995-01-25	20 →	112233 →
3	112233	Kamel Mohamed	Cairo	M	5000.00	1980-05-10	10 →	NULL →

```
304 -- START TRANSACTION;
305
306 -- INSERT INTO Employee
307 -- (SSN, Name, Address, Gender, Salary, BD, DNO, Superssn)
308 -- VALUES
309 -- (102008, 'Hany Ali', 'Cairo', 'M', 2800, '1990-10-10', 10, 112233);
310
311 -- COMMIT;
312
313 START TRANSACTION;
314
315 UPDATE Employee
316 SET Salary = Salary + 500
317 WHERE SSN = 102008;
318
319 ROLLBACK;
320
321
322 START TRANSACTION;
323
324 UPDATE Employee
325 SET Salary = Salary + 500
326 WHERE SSN = 102008;
327
328 SAVEPOINT SalaryUpdate;
329
330 UPDATE Employee
331 SET Address = 'Giza'
332 WHERE SSN = 102008;
333
334 ROLLBACK TO SalaryUpdate;
335
336 COMMIT;
```

line 322, column 1, location 6238

Query 1

Query 2

Query 3

Query 6 OK

```
154
155
156
157
158 INSERT INTO Employee
159 (SSN, Name, Address, Gender, Salary, BD, DNO, Superssn)
160 VALUES
161 (102672, 'Abdullah Farag', 'Bajour', 'A', 150000,
162  '2000-01-01', 30, 112233);|
```



line 162, column 28, location 3840

No limit



Beautify



Query 1 OK: 1 rows affected

```

154
155
156 SELECT
157     Dep.Name AS Dependent_Name,
158     Dep.Gender
159 FROM Dependences Dep
160 JOIN Employee E
161     ON Dep.ESSN = E.SSN
162 WHERE Dep.Gender = 'F'
163 AND E.Gender = 'F'
164
165 UNION
166
167 SELECT
168     Dep.Name AS Dependent_Name,
169     Dep.Gender
170 FROM Dependences Dep
171 JOIN Employee E
172     ON Dep.ESSN = E.SSN
173 WHERE Dep.Gender = 'M'
174 AND E.Gender = 'M';

```



line 174, column 20, location 4011

No limit



Beautify



Run Current

	Dependent_Name	Gender
1	Omar Kamel	M